Downloaded from http://journals.lww.com/indianjurol by BhDMf5ePHKav1zEoum1tQfN4a+kJLhEZgbsIHo4XMi0hCywCX1AWnYQp/IlQrHD3i3D0OdRyi7TvSFl4Cf3VC4/OAVpDDa8K2+Ya6H515kE= on 10/01/2023

Case Report

Elevated prostate-specific antigen in Paget's disease

Shrikant Vasantrao Solav*, Manohar Inamdar[1], Shailendra V. Savale[1]

Department of Nuclear Medicine, SPECTLAB Nuclear Medicine Services, Pune, [1]Department of Medicine, Ashwini Hospital, Akluj, Solapur, Maharashtra, India

*E-mail: drsolav1@gmail.com

ABSTRACT

A 70-year-old man diagnosed with acinar adenocarcinoma prostate (Gleason score 4 + 4 = 8) underwent bone scintigraphy due to high serum prostate-specific antigen level (224 ng/mL; normal <4 ng/mL). Bone scan revealed moderately increased diffuse tracer uptake in the right hemipelvis, while the rest of the scan appeared normal, leading to suspicion of Paget's disease. Gallium-68-prostate-specific membrane antigen (PSMA) positron emission tomography–computed tomography scan (CT) showed a PSMA-avid lesion in the left lobe of the prostate, with no PSMA uptake in the areas of diffuse sclerosis in the right hemipelvis, as seen on CT. This confirmed the diagnosis of Paget's disease.

INTRODUCTION

Paget's disease of the bone is a developmental disorder that affects various bones, leading to abnormal osteoblastic activity detectable on conventional bone scans that vary in intensity depending on the stage of the disease. While many patients remain asymptomatic, a few present with pain, nerve compression, or pathologic factures.

The most commonly involved bones in Paget's disease are the pelvis, femur, spine, skull, and tibia. The enhanced skeletal uptake of radionuclides reflects increased bone formation and blood flow, enabling the detection of Paget's disease before X-ray findings can be appreciated. Pelvic involvement is often asymmetric, with either hemipelvis showing avid tracer concentration on bone scans.

Serum prostate-specific antigen (PSA) is a tumor marker used for the evaluation and follow-up of prostatic carcinoma. Patients with PSA level ≥20 ng/mL have a higher risk of bone metastasis; therefore, bone scintigraphy is highly recommended for these patients. Although bone scintigraphy with Technetium99m-methylene diphosphonate (MDP) is highly sensitive for detecting osteoblastic skeletal metastatic lesions, it has limitations due to suboptimal specificity.

Gallium-68 prostate-specific membrane antigen positron emission tomography–computed tomography scan (Ga68-PSMA PET/CT) is a highly sensitive technique for detecting skeletal involvement in prostatic carcinoma. This case report highlights that caution is required when interpreting bone scans in cases of prostatic carcinoma with elevated PSA levels and evaluates the role of Ga68-PSMA PET/CT in excluding skeletal metastasis.

CASE REPORT

A 70-year-old man presented with symptoms of frequency and hesitancy. Per rectal examination revealed a hard prostate, and sonography showed an enlarged prostate. Serum PSA level was 224 ng/mL (normal <4 ng/mL). Prostatic biopsy confirmed the presence of acinar adenocarcinoma with a Gleason score of 4 + 4 = 8. Furthermore, a bone scan was performed using technetium99m-MDP, which revealed moderately increased tracer uptake in the right

Access this article online	
Quick Response Code:	Website: www.indianjurol.com
	DOI: 10.4103/iju.iju_218_23

This is an open access journal, and articles are distributed under the terms of the Creative Commons Attribution-NonCommercial-ShareAlike 4.0 License, which allows others to remix, tweak, and build upon the work non-commercially, as long as appropriate credit is given and the new creations are licensed under the identical terms.

For reprints contact: WKHLRPMedknow_reprints@wolterskluwer.com

Received: 08.06.2023, **Revised:** 01.08.2023,

Accepted: 16.09.2023, **Published:** 29.09.2023

Financial support and sponsorship: Nil.

Conflicts of interest: There are no conflicts of interest.

Downloaded from http://journals.lww.com/indianjurol by BhDMf5ePHKav1zEoum1tQfN4a+kJLhEZgbsIHo4XMi0hCywCX1AWnYQp/IlQrHD3i3D0OdRyi7TvSFl4Cf3VC4/OAVpDDa8K2+Ya6H515kE= on 10/01/2023

hemipelvis [Figure 1]. However, the left hemipelvis and the rest of the skeleton appeared normal, thereby raising suspicion of Paget's disease. Ga68-PSMA PET/CT was performed on a separate day, which showed increased PSMA ligand uptake in the prostate, normal radiotracer distribution in the entire skeleton, and no increased uptake in the right hemipelvis. Corresponding CT images showed diffuse sclerosis of the right hemipelvis, with subtle areas of osteopenia and lysis interspersed within [Figure 2]. This confirmed the diagnosis of Paget's disease.

DISCUSSION

Skeletal involvement in prostatic carcinoma is commonly observed when PSA levels rise beyond 100 ng/mL, with a positive predictive value ranging between 50% and 75%.[1] PSA is organ specific and not disease specific. PSA levels are elevated in benign prostate enlargement, prostatitis, digital manipulation, or after recent ejaculation. In addition, PSA levels may be elevated in benign and malignant conditions, and up to 30%–40% of patients with prostate cancer may have PSA levels within the normal range.[2] Thus, elevated PSA alone does not translate into prostatic carcinoma-specific mortality and setting a cutoff value of >20 ng/mL to indicate high-risk cancer has been challenged by contradictory results.[3]

Paget's disease may coexist with prostatic carcinoma, often remaining silent and incidentally diagnosed. However, the lesions of Paget's disease of the bone must not be misinterpreted as metastasis.[4] Paget's disease is characterized by regional bone resorption followed by abnormal excessive remodeling.[5]



Figure 1: Whole-body bone scan shows a diffuse mild-to-moderate increase in radiotracer uptake (Tc-99m-MDP) in the right hemipelvis in the anterior (a) and posterior (b) views

Three phases of Paget's disease include the lytic phase, osteoblastic phase, and mixed phase. The osteoblastic phase is associated with increased tracer uptake on bone scan due to sclerosis, causing coarse trabecular and cortical thickening. The osteolytic phase shows a predominant lytic pattern and relatively less osteoblastic activity on bone scans. The intensity of tracer uptake in the mixed phase depends on the dominance of the two phases. Typically, it involves the skull, pelvic bones, and spine.[6]

Various biochemical markers are used for diagnosis, including serum alkaline phosphatase, urinary hydroxyproline, serum osteocalcin, and pyridinium cross-link.[7]

The important causes of osteosclerosis that need to be differentiated from Paget's disease include fibrous dysplasia (characterized by bowing deformity and ground glass opacities), renal osteodystrophy (subperiosteal and subchondral bone resorption), metastasis, myelofibrosis-mastocytosis (both associated with hepatosplenomegaly), fluorosis (ligamentous ossifications), and mastocytosis-tuberous sclerosis (both associated with focal radiodensities).

Ga68-PSMA PET/CT has demonstrated higher sensitivity and specificity in detecting metastatic skeletal lesions than bone scans.[8]

PSMA is expressed 100–1000 times higher in prostatic carcinoma. However, certain benign lesions may exhibit low-grade PSMA ligand uptake, including conditions such as osteoarthritis, degenerative changes, fractures, and fibrous dysplasia. In addition, Paget's disease, liver hemangiomas, bone hemangiomas, splenosis, and sarcoidosis may also show low-grade PSMA ligand uptake.[9] Correlation with morphologic changes on CT, serum PSA levels, and follow-up imaging may help overcome these diagnostic challenges.

In a case reported by Froehner *et al*., Ga68-PSMA PET/CT was found to produce a false-positive result in Paget's disease.[10] The scan showed diffuse increased uptake of both Tc MDP (bone scan) and Ga68 PSMA uptake in the left hemipelvis in a prostatic carcinoma patient with a PSA level of 6.7 ng/mL. This case report was supported by histology obtained from the left acetabulum, which confirmed the presence of Paget's involvement.

Another case report describes metastatic carcinoma of the prostate mimicking Paget's disease.[11]

Downloaded from http://journals.lww.com/indianjurol by BhDMf5ePHKav1zEoum1tQfN4a+kJLhEZgbsIHo4XMi0hCywCX1AWnYQp/IlQrHD3i3D0OdRyi7TvSFl4Cf3VC4/OAVpDDa8K2+Ya6H515kE= on 10/01/2023

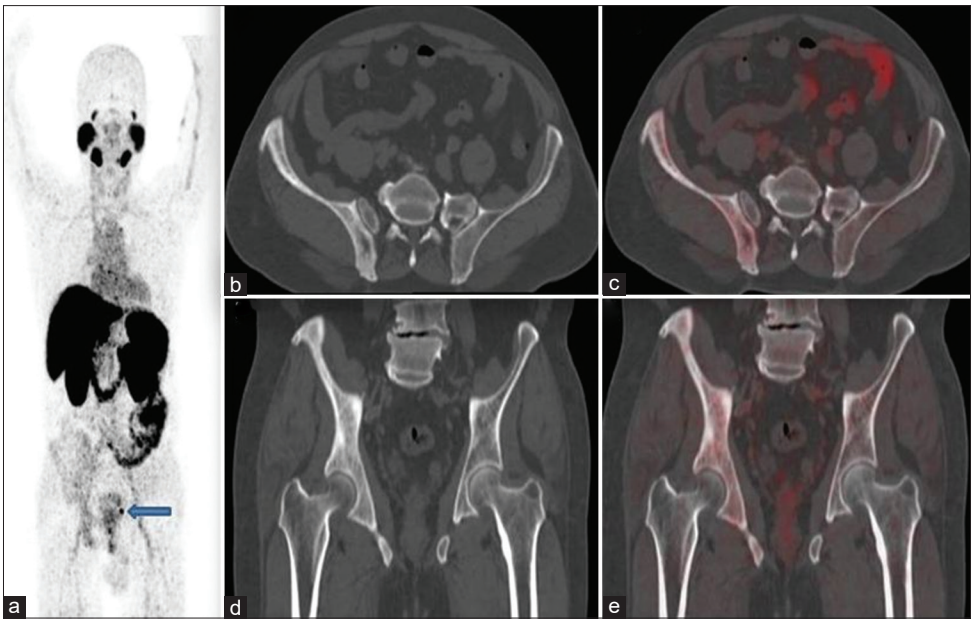


Figure 2: Gallium-68 prostate-specific membrane antigen (PSMA) positron emission tomography–computed tomography scan- MIP image (a) shows a focus of PSMA ligand uptake in the left lobe of the prostate (arrow). CT (b, d) and PET CT fusion images (c, e) of the pelvic bones do not show any evidence of abnormally increased PSMA ligand uptake within the sclerotic areas of the right hemipelvis

CONCLUSION

Elevated PSA in prostatic carcinoma raises a concern for skeletal involvement. In cases where a bone scan shows isolated hemipelvis involvement, it is important not to misinterpret it as metastasis. Ga68-PSMA PET/CT scan can be used to exclude the presence of skeletal metastasis in such situations.

Declaration of the patient's consent

The authors certify that they have obtained all appropriate patient consent forms. In the form, the patient has given consent for his images and other clinical information to be reported in the journal. The patient understands that his name and initials will not be published and due efforts will be made to conceal his identity, but anonymity cannot be guaranteed.

Acknowledgment

The authors would like to thank the team of technologists and associates for helping in carrying out this work.

REFERENCES

1. Stattin K, Sandin F, Bratt O, Lambe M. The risk of distant metastases and cancer specific survival in men with serum prostate specific antigen values above 100 ng/ml. J Urol 2015;194:1594-600.
2. Partin AW, Carter HB, Chan DW, Epstein JI, Oesterling JE, Rock RC, *et al.* Prostate specific antigen in the staging of localized prostate cancer: Influence of tumor differentiation, tumor volume and benign hyperplasia. J Urol 1990;143:747-52.
3. Garg H, Seth A, Kumar R. Raised prostate-specific antigen alone may not be a true predictor in high-risk prostate cancer: A retrospective cohort analysis. Indian J Urol 2022;38:22-8.
4. De Coster L, Sciot R, Everaerts W, Gheysens O, Verscuren R, Deroose C, *et al.* Fibrous dysplasia minicking bone metastasis on (68)Ga-PSMA PET/MR. Eur J Nucl Med Mol Imaging 2017;44:1607-8.
5. Resnick D. Paget disease of bone: Current status and a look back to 1943 and earlier. AJR Am J Roentgenol 1988;150:249-56.
6. Smith SE, Murphey MD, Motamedi K, Mulligan ME, Resnik CS, Gannon FH. From the archives of the AFIP. Radiologic spectrum of Paget disease of bone and its complications with pathologic correlation. Radiographics 2002;22:1191-216.
7. Eastell R. Biochemical markers of bone turnover in Paget's disease of bone. Bone 1999;24:49S-50S.
8. Pyka T, Okamoto S, Dahlbender M, Tauber R, Retz M, Heck M, *et al.* Comparison of bone scintigraphy and (68)Ga-PSMA PET for skeletal staging in prostate cancer. Eur J Nucl Med Mol Imaging 2016;43:2114-21.
9. Hofman MS, Hicks RJ. How we read oncologic FDG PET/CT. Cancer Imaging 2016;16:35.
10. Froehner M, Toma M, Zöphel K, Novotny V, Laniado M, Wirth MP. PSMA-PET/CT-positive Paget disease in a patient with newly diagnosed prostate cancer: Imaging and bone biopsy findings. Case Rep Urol 2017;2017: Article ID 1654231, 3 pages, 2017
11. Fig LM, Gross MD. Metastatic prostate carcinoma mimicking Paget's disease on bone imaging. Clin Nucl Med 1989;14:777-8.

How to cite this article: Solav SV, Inamdar M, Savale SV. Elevated prostate-specific antigen in Paget's disease. Indian J Urol 2023;39:328-30.